

Exact holding region of the Tuan–Thuong weighted Shapiro cyclic inequality for $n = 5$ and $n = 7$

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Abstract

Tuan and Thuong [2] introduced the weighted Shapiro cyclic inequality

$$P_{n,p,q}(x) := \sum_{i=0}^{n-1} \frac{x_i}{p x_{i+1} + q x_{i+2}} \geq \frac{n}{p+q}, \quad x_{i+n} = x_i, \ x_i \geq 0,$$

and posed, as Open Question (b), the problem of finding —for each fixed n —sufficient conditions on (p, q) for the inequality to hold. Both sides scale by the same factor under $(p, q) \mapsto c(p, q)$, so we normalize $p + q = 1$ (bound n) and study the **holding region**

$$H_n = \{p \in (0, 1) : P_{n,p,1-p}(x) \geq n \forall x \text{ admissible}\} = \{p : m_n(p) \geq n\}, \quad m_n(p) = \inf_x P_{n,p,1-p}(x),$$

where *admissible* means $x_i \geq 0$ and $p x_{i+1} + q x_{i+2} > 0$ for all i . Our result is a stronger, exact answer to that question for $n = 5, 7$.

We determine H_5 and H_7 **completely and rigorously**:

$$H_5 = (0, 1), \quad H_7 = (0, a_7] \cup [b_7, 1),$$

where a_7, b_7 are the two roots in $(0, 1)$ of an explicit irreducible degree-15 integer polynomial. Numerically,

$$a_7 \approx 0.21427352091 \in (1/5, 1/4), \quad b_7 \approx 0.32862767792 \in (1/4, 1/3).$$

In particular $H_5 = (0, 1)$: the inequality holds for **all** $p \in (0, 1)$ at $n = 5$, so the sufficient interval $[(5 - \sqrt{5})/10, (5 + \sqrt{5})/10]$ proved by Tuan–Thuong is strict, not exact. For $n = 7$ the first failure band appears: $m_7(p) < 7$ exactly on the open interval (a_7, b_7) , with $m_7(p) = \min\{7, P_{S_2}^{\text{curve}}(p)\}$.

Every face of the simplex (dihedral orbit of an independent set of the cycle C_n) is settled by exact algebraic certificates —resultant elimination, rational Sturm sequences, a positive-coefficient crossing resultant, a negative-determinant saddle certificate, and the Nowosad–Yamagami interior-uniqueness theorem —each independently reconstructed and verified (exact division, remainder zero) rather than taken on trust from any CAS black box. This gives a complete, computer-assisted but fully rigorous answer to Open Question (b) of Tuan–Thuong for $n = 5$ and $n = 7$.

1 Introduction

The classical Shapiro cyclic inequality $\sum x_i/(x_{i+1} + x_{i+2}) \geq n/2$ is false in general; the first counterexample is $n = 20$ (Lighthill, 1956). The modern picture is that the inequality holds for even $4 \leq n \leq 12$ and odd $3 \leq n \leq 23$, and fails for even $n \geq 14$ and odd $n \geq 25$ (Ando [5] gives a proof of the Shapiro inequality and a broad classification framework for related cyclic inequalities). Drinfeld [1] determined the asymptotic constant. Tuan and Thuong [2], under the title *On an Extension of Shapiro’s Cyclic Inequality*, introduced the two-parameter weighted form above, completely classified $n = 4$, gave the sufficient interval $[(5 - \sqrt{5})/10, (5 + \sqrt{5})/10]$ for $n = 5$, and proved failure for every even $n \geq 4$ when $p < q$. The exact (p, q) -region for odd $n \geq 5$ was left open.

We normalize $p + q = 1$ (bound n) and write $P_{n,p} := P_{n,p,1-p}$. The value function $m_n(p) = \inf_x P_{n,p}(x)$ is continuous on $(0, 1)$ (Lemma 2.1 below); the holding region is $H_n = \{p : m_n(p) \geq n\}$. The two smallest odd cases are settled here.

Main results.

- **Theorem 1 ($n = 5$, rigorous).** $H_5 = (0, 1)$; equivalently $m_5(p) \equiv 5$ on $(0, 1)$. The Tuan–Thuong sufficient interval is strict: $[(5 - \sqrt{5})/10, (5 + \sqrt{5})/10] \subsetneq (0, 1) = H_5$.
- **Theorem 2 ($n = 7$, rigorous).** $H_7 = (0, a_7] \cup [b_7, 1)$ where $a_7 < b_7$ are the two $(0, 1)$ -roots of the irreducible degree-15 integer polynomial F in §4, pinned down by exact Sturm isolation as $a_7 \in (0.214273512, 0.214273525)$ and $b_7 \in (0.328627665, 0.328627686)$ (so a_7, b_7 are defined algebraically, not as experimental decimals). Here $H_n \subset (0, 1)$ throughout, so the endpoints 0 and 1 are excluded by the notation. Equivalently $m_7(p) = \min\{7, P_{S_2}^{\text{curve}}(p)\}$, and $m_7(p) < 7 \iff p \in (a_7, b_7)$.

Theorems 1-2 give the exact answer to Open Question (b) of Tuan-Thuong for $n = 5, 7$. The appearance of the first failure band at $n = 7$ (absent at $n = 5$) is a boundary-induced transition: since $\frac{5-\sqrt{5}}{10} \approx 0.2764 < a_7$ and $b_7 < \frac{5+\sqrt{5}}{10}$, the band (a_7, b_7) satisfies $(\frac{5-\sqrt{5}}{10}, b_7) \subset (a_7, b_7)$, so the $n = 5$ sufficient condition provably does **not** extend to $n = 7$.

2 Preliminaries

Lemma 2.1 (continuity of m_n). For each fixed odd n , $m_n(p) = \inf_{x \in \Delta_n} P_{n,p}(x)$ is continuous on $(0, 1)$.

Proof. Extend $P_{n,p}$ to the closed simplex $\Delta_n = \{x \geq 0 : \sum x_i = 1\}$ with values in $\mathbb{R} \cup \{+\infty\}$, setting a summand $x_i/(p x_{i+1} + q x_{i+2})$ to $+\infty$ whenever $x_i > 0$ and the denominator is 0 (and $0/0 := 0$). Each summand is lower-semicontinuous (lsc) in (x, p) on $\Delta_n \times (0, 1)$ —it is continuous wherever the denominator is positive and jumps to $+\infty$ only when a denominator vanishes under a positive numerator —so P is jointly lsc. Let $\Omega := \{(x, p) : P_{n,p}(x) < +\infty\}$ be the **admissible interior** (all relevant denominators > 0); on Ω , P is C^∞ (rational). Since $P_{n,p}(\mathbf{1}) = n$ for every p , we have $m_n(p) \leq n$.

Attainment (coercivity). For fixed p , the sublevel $\{x \in \Delta_n : P_{n,p}(x) \leq n\}$ is closed (lsc) in the compact Δ_n , hence compact, and nonempty (it contains $\mathbf{1}$); so the infimum is attained at some x^* with $P_{n,p}(x^*) = m_n(p) \leq n < +\infty$, i.e. $x^* \in \Omega_p$. This is the boundary blow-up / coercivity: $P_{n,p}(x) \rightarrow +\infty$ at every *infeasible* boundary point (one with $x_i > 0$ but $p x_{i+1} + q x_{i+2} \rightarrow 0$), uniformly in p on compact subintervals (since $p, q \geq \alpha > 0$), so no minimizer can sit on the infeasible boundary; the minimizer lies in the admissible interior.

Continuity. Let $p_k \rightarrow p$ and let x_k attain $m_n(p_k)$ (so $P_{n,p_k}(x_k) \leq n$). By compactness of Δ_n , after passing to a subsequence $x_k \rightarrow x^*$; by joint lsc, $m_n(p) \leq P_{n,p}(x^*) \leq \liminf_k m_n(p_k)$ (lower semicontinuity). Conversely, let x^* attain $m_n(p)$; then $x^* \in \Omega_p$, and since Ω is open and P is jointly continuous on Ω , $P_{n,p_k}(x^*) \rightarrow P_{n,p}(x^*) = m_n(p)$, whence $m_n(p_k) \leq P_{n,p_k}(x^*)$ gives $\limsup_k m_n(p_k) \leq m_n(p)$ (upper semicontinuity). Thus m_n is continuous. $\square$

(Equivalence of finiteness and admissibility: $P_{n,p}(x) < +\infty$ iff every denominator $p x_{i+1} + q x_{i+2}$ is strictly positive. Necessity is the lsc blow-up above. For sufficiency, if no denominator vanishes then each summand is finite and continuous, so P is finite; the only way a denominator can vanish while $x \in \Delta_n$ is a maximal consecutive zero-block producing a positive numerator over a zero denominator —exactly the $+\infty$ case. Hence $\Omega = \{P < +\infty\}$ is precisely the admissible domain, and the boundary-inheritance recursion of Lemma 2.3 ranges only over viable independent zero-sets.)

Consequently $H_n = \{p : m_n(p) \geq n\}$ is closed in $(0, 1)$, so its complement (the failure set) is open —a union of open bands.

Faces and supports. P is degree-0 homogeneous in x , so we minimize on the simplex $\sum x_i = 1$. A support $S = \{i : x_i > 0\}$ is **viable** only if its zero-set contains no two cyclically adjacent indices (otherwise a denominator vanishes). Hence viable zero-sets are **independent sets of the cycle C_n** . By degree-0 homogeneity and Euler's theorem, at an interior minimizer on a support, $\nabla_S P = 0$ (the Lagrange multiplier λ in $\nabla P = \lambda \mathbf{1}$ vanishes since $\sum x_i \partial_i P = 0$). Each orbit's infimum is either an interior KKT point (a **stable** support) or is attained on its boundary (degenerating to a smaller orbit).

Gap structure. A zero-set decomposes the cycle into gaps; up to dihedral symmetry a zero-set is a **gap word** $(g_1, \dots, g_m)$ with $\sum g_i = n$ and every $g_i \geq 2$. The boundary of an orbit is the union of orbits obtained by enlarging the zero-set (adding one more zero while keeping the set independent).

> **Lemma 2.3 (boundary inheritance).** For a precise-support stratum $\Sigma_Z := \{x \in \Delta_n : \{i : x_i = 0\} = Z\}$ with Z an independent set of C_n ,

>

$$\inf_{\Sigma_Z} P = \min \left\{ \text{interior local-minimum KKT values on } \Sigma_Z, \min_{Z' \supsetneq Z, Z' \text{ independent}} \inf_{\Sigma_{Z'}} P \right\},$$

> with the convention $\inf_{\Sigma_{Z'}} P = +\infty$ when Z' is non-viable (contains adjacent zeros).

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> *Proof.* Extend P to the closed face $\overline{\Sigma_Z} = \{x \in \Delta_n : \text{supp}(x)^c \supseteq Z\}$ as in Lemma 2.1 (lsc, $+\infty$ on infeasible points); $P < +\infty$ iff every denominator is strictly positive, so the admissible domain equals the union of precise-support strata $\bigcup_{Z' \supseteq Z, Z' \text{ ind.}} \Sigma_{Z'}$. Take a minimizing sequence $x_k \in \Sigma_Z$ with $P(x_k) \rightarrow \inf_{\Sigma_Z} P$; by compactness of Δ_n , after passing to a subsequence $x_k \rightarrow x^* \in \overline{\Sigma_Z}$. If $\text{supp}(x^*)^c = Z$ (the support did not shrink), then $x^* \in \Sigma_Z$ and, being an interior minimizer on Σ_Z , satisfies $\nabla_{\Sigma_Z} P = 0$ (Euler, $\lambda = 0$) —an interior KKT value. Otherwise the zero-set enlarged to an independent $Z' \supsetneq Z$, so $x^* \in \Sigma_{Z'}$ and $P(x^*) \geq \inf_{\Sigma_{Z'}} P$ (lsc). If the enlargement ever creates two cyclically adjacent zeros, take a maximal consecutive zero-block: the positive component just before it has a numerator bounded away from 0 while its denominator (a sum involving a zero neighbour) tends to 0, so $P \rightarrow +\infty$ —hence non-viable Z' contributes $+\infty$ and never attains the infimum. Conversely, each viable child Z' is approachable from Σ_Z by lifting its new zero to $\varepsilon \downarrow 0$ (independence keeps all denominators > 0 for small ε), so $\inf_{\Sigma_{Z'}} P$ is indeed inherited from below. Taking the minimum over the interior KKT values and all viable children gives the displayed formula. $\square$

This is the stratum recursion used throughout §3–§4: $\inf_{S_0} P = \min\{7, \inf_{S_1} P, \dots\}$, etc. (Here S_0 denotes the *open* full-support stratum $\{x_i > 0 : \sum x_i = 1\}$, distinct from the closed simplex Δ_n ; the minimizer on Δ_n is either the interior point **1** or lies on a viable boundary stratum, which is exactly what the lemma formalises.) Yamagami [4, §5, pp.524–527] likewise notes that interior Hessian uniqueness does not control the boundary, which must be analysed separately —the stratum recursion is that analysis.

Nowosad–Yamagami theorem (Nowosad [3], CPAM 21 (1968), Thm 1.8; Yamagami [4], Proc. AMS 118 (1993), 521–527, Thm 1, Lemma 3, Cor. 5). We state the finite-dimensional specialisation we need, in the commutative C^* -algebra $A = \mathbb{C}^n$ (entrywise product, state $\varphi = \sum_i$, self-adjoint part $A_s = \mathbb{R}^n$). For a linear $T : A \rightarrow A$ write $X_T(y) = \varphi(y^{-1}T(y)) = \sum_i T(y)_i / y_i$ on A_{++} .

(NY-1) *Constancy on the closed subgroup* (Yamagami Thm 1, restating Nowosad Thm 1.8). For $a \in A_{++}$, let $[a] \subset A_s^\times$ denote Yamagami’s closed subgroup consisting of those self-adjoint invertibles x for which both x and x^{-1} are norm-limits of Laurent polynomials in a ; it contains a (so it is *not* the geometric segment $\{(1-t)\mathbf{1} + ta : 0 \leq t \leq 1\}$, and in general is neither compact nor contained in the positive cone). If T is $*$ -preserving ($T(x^*) = T(x)^*$) and X_T has local maxima at **1** and at $a \in A_s \setminus \mathbb{R}\mathbf{1}$, then X_T is **constant on** $[a]$. In finite dimension $a^t = \exp(t \log a) \in [a]$ for every $t \in \mathbb{R}$ (by polynomial interpolation on the finite spectrum of a); constancy gives $\frac{d^2}{dt^2} X_T(a^t)|_{t=0} = 0$, so the tangent direction $\log a$ (mod the scale direction **1**, nonzero when a is non-scalar) lies in $\ker \nabla^2 X_T(\mathbf{1})$. Consequently, if $\ker \nabla^2 X_T(\mathbf{1}) = \text{span}\{\mathbf{1}\}$, no such non-scalar a exists: X_T has at most one isolated interior local maximum, up to the scalar ray.

(NY-2) *Spectral non-degeneracy* (Yamagami Lemma 2 + Lemma 3; Cor. 5). Yamagami’s Hessian formula (Lemma 2) gives, for $P = f_S$ with $S\mathbf{1} = \mathbf{1}$ ($s = 1$), the Hessian on $\mathbf{1}^\perp$ diagonalised by the Fourier modes; the same Fourier calculation as in Cor. 5 yields the eigenvalue $\mu_k = 2|\lambda_k|^2 - 2\text{Re } \lambda_k$ (identically $2(1 - \cos \theta_k)[2q \cos \theta_k + 2p^2 - 3p + 2]$, verified below), where $\lambda_k = p\omega^k + q\omega^{2k}$ ($k \neq 0$). Cor. 5 itself states the **interior-disk** condition $|\lambda_k - \frac{1}{2}| < \frac{1}{2}$ for the f_S -*maximum* case ($\mu_k < 0$, the negative Hessian $s(S + S^\top) - 2S^\top S$ being positive semidefinite); our $P = f_S$ *minimum* case is the **symbol-dual**: $\mu_k > 0$ iff $|\lambda_k - \frac{1}{2}| > \frac{1}{2}$ (exterior disk), equivalently $\text{sign } \mu_k = \text{sign}(|\lambda_k - \frac{1}{2}|^2 - \frac{1}{4})$ (verified in `verify_ny_spectral.py`). Thus $\mu_k > 0$ for all $k \neq 0$ makes **1** a **strict local minimum** with $\ker \nabla^2 P(\mathbf{1}) = \text{span}\{\mathbf{1}\}$, and by (NY-1) the **unique** interior local-minimum ray.

> **Lemma 2.2 (interior uniqueness, finite-dimensional N–Y).** Let $S = pC + qC^2$ ($p, q > 0$, $p + q = 1$, n odd) and $P(x) = \sum_{i=0}^{n-1} x_i / (Sx)_i$ on the positive cone. If $\nabla^2 P(\mathbf{1}) \succeq 0$ with $\ker \nabla^2 P(\mathbf{1}) = \text{span}\{\mathbf{1}\}$ (equivalently every $\mu_k > 0$; for $n = 5, 7$ this follows uniformly in p from $\Delta_n < 0$, proved in §3.1 and §4.1), then **1** is the **unique interior local-minimum ray** of

P on the full-support face. The inequality $P \geq n$ on the whole face is *not* a consequence of this lemma; it requires a separate boundary analysis (Yamagami [4], §5, pp.524–527), performed here by stratum recursion (boundary inheritance) in §3.4 ($n = 5$) and §4.5 ($n = 7$).

Proof. (1) **S is invertible.** S is circulant with Fourier eigenvalues $\lambda_k = \omega^k(p + q\omega^k)$; $\lambda_k = 0$ would force $\omega^k = -p/q$ of modulus 1, hence $p = q$ and $\omega^k = -1$, impossible for odd n . (2) **$S\mathbf{1} = \mathbf{1}$ and $S^\top \mathbf{1} = \mathbf{1}$** (each column sums to $p + q = 1$), hence $S^{-1}\mathbf{1} = \mathbf{1}$. (3) **$T = -S^{-1}$ is $*$ -preserving, not self-adjoint.** S is real, hence so is S^{-1} , so $T(x^*) = T(x)^*$ (complex conjugation commutes with the real matrix T). This is the structure N-Y requires. (S itself is *not* self-adjoint — $S^\top = pC^{-1} + qC^{-2} \neq S$ for $n \geq 3$ — and self-adjointness of T is neither needed nor true in general; only $*$ -preservation is.) (4) **Sign bridge.** The linear map $S : A_{++} \rightarrow S(A_{++})$ is a diffeomorphism onto its image (by (1), S is invertible); $S(A_{++})$ is open in $\mathbb{R}^n$ and contains a neighbourhood of $S\mathbf{1} = \mathbf{1}$. Since $P(x) = \sum_i (S^{-1}y)_i / y_i = X_{S^{-1}}(y)$ and $X_{-S^{-1}} = -X_{S^{-1}}$ by linearity of φ , a local minimum of P at $\mathbf{1}$ corresponds to a local maximum of $X_{-S^{-1}}$ at $\mathbf{1}$ *within* $S(A_{++})$; because $S(A_{++})$ is a neighbourhood of $\mathbf{1}$, this is also a local maximum in A_{++} , with $X_{-S^{-1}}(\mathbf{1}) = -n$. (We do not claim S^{-1} preserves the positive cone — only the local correspondence at $\mathbf{1}$ is needed.) (5) **Hessian transfer (congruence).** Differentiating $X_T(y) = -P(S^{-1}y)$ twice at $y = \mathbf{1}$ (with $T = -S^{-1}$) gives, in the same coordinate basis, the **congruence**

$$\nabla_y^2 X_T(\mathbf{1}) = -S^{-\top} \nabla_x^2 P(\mathbf{1}) S^{-1}.$$

Since $S\mathbf{1} = \mathbf{1}$, this congruence flips the inertia sign ($H \succeq 0 \iff \nabla_y^2 X_T(\mathbf{1}) \preceq 0$) and preserves the kernel: $\ker \nabla_y^2 X_T(\mathbf{1}) = S \ker \nabla_x^2 P(\mathbf{1}) = \text{span}\{\mathbf{1}\}$. Thus $\mathbf{1}$ is a non-degenerate strict local maximum of X_T modulo scale — the Hessian hypothesis of (NY-1). (6) **Uniqueness via (NY-1).** Suppose P had a second non-scalar interior local-minimum ray x^* . Then $y^* = Sx^*$ is a second local maximum of X_T ; by (NY-1) X_T is constant on the closed subgroup $[y^*]$, so the curve $(y^*)^t = \exp(t \log y^*) \in [y^*]$ has $\frac{d^2}{dt^2} X_T((y^*)^t)|_{t=0} = 0$, placing the non-scale tangent $\log y^*$ in $\ker \nabla_y^2 X_T(\mathbf{1})$. This contradicts $\ker \nabla_y^2 X_T(\mathbf{1}) = \text{span}\{\mathbf{1}\}$ from (5). Hence no second interior local minimum exists: $\mathbf{1}$ is the unique one. (7) This is an *interior* statement on the full-support face; the global infimum may lie on a boundary stratum and is closed separately by stratum recursion. $\square$

Uniform Hessian spectrum. At $x = \mathbf{1}$, the Hessian of $P_{n,p}$ on $\mathbf{1}^\perp$ is diagonalized by the Fourier modes $\theta_k = 2\pi k/n$ ($k = 1, \dots, n-1$):

$$\mu_k = 2(1 - \cos \theta_k) [2q \cos \theta_k + 2p^2 - 3p + 2].$$

The most dangerous mode is $k = (n-1)/2$; writing $c_n = -\cos(\pi/n)$, the bracket there is a quadratic in p with discriminant $\Delta_n = 4c_n^2 - 4c_n - 7$. For $n = 5, 7$, $\Delta_n < 0$ (proved rigorously in §3.1 and §4.4), so the bracket is strictly positive for all $p \in (0, 1)$, hence every $\mu_k > 0$: the uniform point is a **strict** local minimum with $\ker \nabla^2 P(\mathbf{1}) = \text{span}\{\mathbf{1}\}$, and Lemma 2.2 applies.

Certificate toolbox. All algebraic assertions are obtained by: (i) eliminate the auxiliary variable(s) via a **resultant**; (ii) count real roots in an interval by an exact rational **Sturm sequence** (recording the sign-variation counts at the endpoints); (iii) certify sign-constancy by a **positive-coefficient** resultant factor (a polynomial with all coefficients of one sign is sign-definite on $(0, \infty)$, hence has no positive root), or by resultant+Sturm on a degeneracy locus; (iv) isolate each real root in a rational interval and certify signs by rational interval arithmetic. Every polynomial below is locally reconstructed and verified (exact division, quotient and remainder recorded, remainder zero) independently of any CAS black box. Code for every certificate is listed in §5.

3 The $n = 5$ classification (Theorem 1, rigorous)

The independence number of C_5 is $\alpha(C_5) = 2$, so viable zero-sets have at most two elements. Up to the dihedral action of C_5 there are exactly **three** orbits:

orbit	zero-set	support	gap word	type
S_0	$\emptyset$	5	—	full (uniform)
S_1	$\{0\}$	4	(5)	1-zero, one-gap $L = 5$
S_2	$\{0, 2\}$	3	(2, 3)	2-zero, one-gap $L = 3$ (AM-GM)

We show $P > 5$ on S_1, S_2 and $P = 5$ only at the uniform point of S_0 ; boundary degeneration then gives $m_5(p) = 5$.

3.1 S_0 —Nowosad–Yamagami

At $x = \mathbf{1}$, $P = 5$. The uniform Hessian spectrum (§2) has $\Delta_5 = 4c_5^2 - 4c_5 - 7$ with $c_5 = -\cos(\pi/5) = -\frac{1+\sqrt{5}}{4}$. Then $4c_5^2 = \frac{3+\sqrt{5}}{2}$ and $4c_5 = -(1+\sqrt{5})$, so

$$\Delta_5 = \frac{3+\sqrt{5}}{2} + (1+\sqrt{5}) - 7 = \frac{3(\sqrt{5}-3)}{2} \approx -1.146 < 0.$$

Hence every $\mu_k > 0$ on $\mathbf{1}^\perp$, so $\nabla^2 P(\mathbf{1}) \succeq 0$ with kernel $\text{span}\{\mathbf{1}\}$. By Lemma 2.2, $\mathbf{1}$ is the **unique interior local-minimum ray** of P on the full-support stratum S_0 for every $p \in (0, 1)$, with value $P(\mathbf{1}) = 5$. This is a *local* statement; the face infimum $\inf_{S_0} P$ is closed by **boundary inheritance** (Lemma 2.3): the minimizer on the closed face $\overline{S_0} = \Delta_5$ is either the interior critical point $\mathbf{1}$ (value 5) or lies on a viable boundary stratum, so $\inf_{S_0} P = \min\{5, \inf_{S_1} P, \inf_{S_2} P\}$.

3.2 S_2 —AM-GM closed form

With $x_0 = x_2 = 0$ and support $\{1, 3, 4\}$,

$$P = \frac{x_1}{q x_3} + \frac{x_3}{p x_4} + \frac{x_4}{q x_1},$$

whose three summands have product $1/(p q^2)$. By AM-GM,

$$P \geq 3(p q^2)^{-1/3} =: M_{5,3}(p), \quad \text{with equality iff } x_1/(q x_3) = x_3/(p x_4) = x_4/(q x_1).$$

The minimum of $M_{5,3}$ over $p \in (0, 1)$ is at $p = 1/3$ (where $p q^2$ is maximized):

$$\min_p M_{5,3} = \frac{3}{\left(\frac{1}{3}\left(\frac{2}{3}\right)^2\right)^{1/3}} = \frac{9}{\sqrt[3]{4}} \approx 5.6696 > 5,$$

rigorously since $9^3 = 729 > 500 = 5^3 \cdot 4$. Hence S_2 **never fails**: $\inf_{S_2} P \geq 9/\sqrt[3]{4} > 5$.

3.3 S_1 —stationary branch, positive-coefficient certificate

This is the only non-closed-form face. Set $x_0 = 0$, $x_4 = 1$ (homogeneity), $a = x_1, b = x_2, c = x_3$, $q = 1 - p$. Then

$$P = \frac{a}{p b + q c} + \frac{b}{p c + q} + \frac{c}{p} + \frac{1}{q a}.$$

The KKT equations $\partial_a P = \partial_b P = \partial_c P = 0$ (Euler, $\lambda = 0$) reduce exactly to

$$p b + q c = q a^2, \quad p c + q = \frac{q^2 a^3}{p}, \quad \boxed{q^3 a^6 - p^3 a^2 - p^2 q = 0} \quad (\text{stationary curve}).$$

(The first two give $c = (q^2 a^3 - p q)/p^2$, $b = q(a^2 - c)/p$; substituting into $\partial_c P = 0$ and cancelling the $p q^2 a^3$ pair yields the curve.)

Dehomogenization. Write $\tilde{P}_r := P_{5,r,1}$ (parameters $p : q = r : 1$); since both sides of the inequality scale identically under $(p, q) \mapsto c(p, q)$, the normalized value at $p = r/(1+r), q =$

$1/(1+r)$ is $(1+r)\tilde{P}_r$. The curve becomes $a^6 - r^3a^2 - r^2 = 0$, which for each $r > 0$ has a **unique positive root** $a(r)$ (with $u = a^2$: $u^3 - r^3u - r^2 = 0$ has $f(0) < 0$, $f \rightarrow +\infty$, and a negative minimum, hence one positive crossing). The branch is a smooth connected curve over $r > 0$.

Positivity of the support point. With $p : q = r : 1$ the recovery formulas are $c = (a^3 - r)/r^2$ and $b = (a^2 - c)/r$. Put $y := a^3/r > 0$; the curve gives $ra^2 = y^2 - 1$. Since $a, r > 0$ we have $ra^2 > 0$, hence $y^2 > 1$ and, as $y > 0$, $y > 1$. Therefore $c = (y - 1)/r > 0$, and $a^2 - c = (y^2 - 1)/r - (y - 1)/r = y(y - 1)/r > 0$, so $b = (a^2 - c)/r > 0$. The eliminated branch thus lies entirely in the positive support.

No-crossing certificate. The degree-0 target is $M := (1+r)\tilde{P}_r - 5$. Let $N(a, r) = \text{num}((1+r)\tilde{P}_r - 5)$ on the branch. Eliminating a :

$$\text{Res}_a(a^6 - r^3a^2 - r^2, N(a, r)) = r^{17} Q_{11}(r),$$

where (coefficients high-to-low)

$$Q_{11}(r) = 1024r^{11} + 3644r^{10} + 6360r^9 + 11580r^8 + 15895r^7 + 28772r^6 + 23892r^5 + 21120r^4 + 9360r^3 + 7680r^2 + 2368r + 1$$

Every coefficient of Q_{11} is positive, so $Q_{11}(r) > 0$ for all $r > 0$. Since $r > 0$ on the branch and Q_{11} is (up to the prefactor r^{17}) the crossing resultant, M **never vanishes** on the stationary curve: $P \neq 5$ there. The sign is constant on the connected branch, and it suffices to evaluate at one point.

Sign certificate at $r = 1$. At $r = 1$ ($p = q = 1/2$) the curve is $a^6 - a^2 - 1 = 0$; its unique positive root is isolated in $a \in (1.150, 1.151)$ ($f(1.150) = -604111/64000000 < 0 < f(1.151) = 354174281094401/10^{15}$, and $f'(a) = 6a^5 - 2a > 0$ on this interval, so the root is unique there). On this interval rational interval arithmetic gives

$$M(1) = \frac{6}{a} + \frac{2}{a^3} + 2a^3 - 9 \in [0.5662, 0.5800] > \frac{1}{2} > 0$$

(the simplification uses $b + c = a^2$, $c + 1 = a^3$ on the branch). Hence $P > 5$ **on the entire S_1 stationary branch**. The boundary $\partial S_1 = S_2$ has $P > 5$ (§3.2), so $\inf_{S_1} P > 5$.

3.4 Conclusion of Theorem 1

The three orbits exhaust all viable supports ($\alpha(C_5) = 2$). **Boundary inheritance:** for each face, P is lsc on the compact face-closure and blows up $(+\infty)$ on its *infeasible* boundary, so its infimum is attained either at an interior critical point or on a viable boundary stratum. Thus

$$\inf_{S_0} P = \min\{P(\mathbf{1}), \inf_{S_1} P, \inf_{S_2} P\} = \min\{5, > 5, > 5\} = 5,$$

using Lemma 2.2 (unique interior local min on S_0 , value 5), $\inf_{S_1} P > 5$ (§3.3), and $\inf_{S_2} P \geq 9/\sqrt[3]{4} > 5$ (§3.2). The recursion $S_0 \rightarrow S_1 \rightarrow S_2 \rightarrow \emptyset$ (each ∂S_k the next viable stratum, S_2 terminal) closes every face. Therefore

$$m_5(p) = 5 \quad \forall p \in (0, 1), \quad H_5 = (0, 1).$$

The Tuan-Thuong interval $[(5 - \sqrt{5})/10, (5 + \sqrt{5})/10] \approx [0.2764, 0.7236]$ is a strict subset of H_5 : sufficient but not exact. ■

4 The $n = 7$ classification (Theorem 2, rigorous)

The independent sets of C_7 fall into five dihedral orbits:

orbit	zero-set	support	gap word	type
S_0	$\emptyset$	7	—	full (uniform)
S_1	$\{0\}$	6	(7)	1-zero, $L = 7$
S_2	$\{0, 2\}$	5	(2, 5)	2-zero, $L = 5$ (main / failing)
S_3	$\{0, 3\}$	5	(3, 4)	2-zero
S_4	$\{0, 2, 4\}$	4	(2, 2, 3)	3-zero, $L = 3$

The boundary of an orbit is obtained by enlarging the zero-set: $\partial S_0 = S_1$, $\partial S_1 = S_2 \cup S_3$ (a second zero is either $\{0, 2\} \sim S_2$ or $\{0, 3\} \sim S_3$), $\partial S_2 \subset S_4$, $\partial S_3 \subset S_4$. Since $\alpha(C_7) = 3$, no viable zero-set has more than three elements; the three-zero orbit is S_4 (unique, gap word (2, 2, 3), the only partition of 7 into three parts ≥ 2), and these **five orbits exhaust all viable supports**. S_4 is *terminal*: adding a fourth zero to $\{0, 2, 4\}$ always creates two cyclically adjacent zeros (positions 1, 3, 5, 6 each border an existing zero), so ∂S_4 is entirely non-viable ($P \rightarrow +\infty$). Thus the boundary recursion $S_0 \rightarrow S_1 \rightarrow \{S_2, S_3\} \rightarrow S_4 \rightarrow \emptyset$ terminates, and $m_7(p) = \min\{\inf_{S_k} P : 0 \leq k \leq 4\}$ is controlled by the five face certificates below.

4.1 Closed forms: S_0, S_4, S_3

- S_0 (**uniform**). $x_i \equiv c$ gives $P = 7$. Write $u_7 = \cos(\pi/7) \in (9/10, 91/100)$ (it is the root of $8u^3 - 4u^2 - 4u + 1 = 0$ in $(0, 1)$; $f(9/10) = -1/125 < 0 < f(91/100)$), and $c_7 = -u_7$ as in §2. The discriminant is $\Delta_7 = 4u_7^2 + 4u_7 - 7 = 4c_7^2 - 4c_7 - 7$; the function $g(u) = 4u^2 + 4u - 7$ is increasing ($g'(u) = 8u + 4 > 0$) with $g(9/10) = -4/25 < 0$ and $g(91/100) < 0$, so $\Delta_7 = g(u_7) < 0$. Hence every uniform-Hessian eigenvalue $\mu_k > 0$, Lemma 2.2 applies, and **1** is the **unique interior local-minimum ray** on S_0 (value 7). This is a *local* statement; the face infimum is closed by **boundary inheritance** (Lemma 2.3): the minimizer on the closed face $\overline{S_0} = \Delta_7$ is either **1** (value 7) or lies on a viable boundary stratum, so $\inf_{S_0} P = \min\{7, \inf_{S_1} P, \inf_{S_2} P, \inf_{S_3} P, \inf_{S_4} P\}$ — *not* pointwise $P \geq 7$ on the open stratum S_0 .
- S_4 — **AM-GM**. With $x = (0, 1, 0, b, 0, c, d)$, $q = 1 - p$, the four summands are $\frac{1}{qb}, \frac{b}{qc}, \frac{c}{pd}, \frac{d}{q}$, whose product is $\frac{1}{pq^3}$. AM-GM gives

$$P_{S_4} \geq 4(pq^3)^{-1/4}, \quad \text{equality iff } \frac{1}{qb} = \frac{b}{qc} = \frac{c}{pd} = \frac{d}{q},$$

i.e. $b = (p/q)^{1/4}, c = (p/q)^{1/2}, d = (q/p)^{1/4}$ (the KKT solution). The minimum over p is at $p = 1/4$ (maximizing pq^3):

$$\min_p P_{S_4} = \frac{16}{3^{3/4}} \approx 7.0190614 > 7 \quad (16^4 = 65536 > 64827 = 7^4 \cdot 3^3).$$

S_4 **never fails**. All finite boundaries of S_2, S_3 lie in S_4 , hence are > 7 .

- S_3 — **exact sign certificate**. With $r = (q/p)^{1/5}$, the KKT solution is $b = r^2, c = r^{-1}, e = r^3, d = r(1 - r^7)$; $d > 0 \iff r < 1 \iff p > 1/2$. For $p \leq 1/2$ (the whole failure band $(a_7, b_7) \subset (0, 1/3)$) there is **no** positive interior stationary point, so $\inf_{S_3} P$ is attained on $\partial S_3 \subset S_4$ and is > 7 . For $p > 1/2$ (so $0 < r < 1$) the stationary value is

$$P_{S_3}^{\text{stat}}(r) = \frac{(1 + r^5)(5 - r^7)}{r^2}.$$

The **exact identity**

$$(1 + r^5)(5 - r^7) - 8r^2 = (1 - r)H(r), \quad H(r) = r^{11} + r^{10} + r^9 + r^8 + r^7 + 2r^6 + 2r^5 - 3r^4 - 3r^3 - 3r^2 + 5r + 5,$$

holds (verified by expansion). For $0 < r < 1$, $H(r) \geq 5 + 5r - 9r^2 \geq 1$: indeed $H(r) - (5 + 5r - 9r^2) = r^2(r^9 + r^8 + r^7 + r^6 + 2r^4 + 2r^3 - 3r^2 - 3r + 6)$ and the bracket is $\geq 6 - 3r^2 - 3r =$

$3(2 - r^2 - r) > 0$ on $(0, 1)$; and $5 + 5r - 9r^2$ is concave with endpoint values 5, 1, hence ≥ 1 . Therefore $P_{S_3}^{\text{stat}} - 8 = \frac{(1-r)H(r)}{r^2} > 0$, i.e.

$$P_{S_3}^{\text{stat}}(r) > 8 > 7 \quad (0 < r < 1).$$

S_3 never fails.

4.2 S_2 (main) —the only failing orbit

$x = (0, 1, b, c, d, 0, e)$, $e = t$. Euler ($\lambda = 0$) gives $d = t^2$, $c = q(q - pt^4)/(p^2t^2)$, $b = q/(pt) - q^2(q - pt^4)/(p^3t^2)$, with the **stationary curve**

$$R(p, t) = q^3 - p^3t^5 - p^2qt^8 = 0.$$

Positive lift. Put $\rho = q/p$ (so $q = p\rho$); $R = 0$ reads $\rho^3 = t^5 + \rho t^8$. As a function of $t > 0$ the left-minus-right side is strictly decreasing ($-5t^4 - 8\rho t^7 < 0$), with value $\rho^3 > 0$ at $t = 0$ and value $-\rho^{5/4} < 0$ at $t = \rho^{1/4}$; hence the unique positive root satisfies $t < \rho^{1/4}$, i.e. $t^4 < \rho$. From $R = 0$, $\rho(\rho - t^4)(\rho + t^4) = \rho^3 - \rho t^8 = t^5$. Therefore

$$c = \frac{\rho(\rho - t^4)}{t^2} > 0, \quad d = t^2 > 0, \quad b = \frac{\rho}{t^2} \left[t - \rho(\rho - t^4) \right] = \frac{\rho}{t^2} \left[t - \frac{t^5}{\rho + t^4} \right] = \frac{\rho^2}{t(\rho + t^4)} > 0,$$

so the stationary point is strictly positive for every $p \in (0, 1)$.

On $R = 0$, $P_{\text{curve}} - 7 = \frac{B(p, t)}{qp^2} + \frac{R(\dots)}{qp^3t^4}$, so $P_{\text{curve}} = 7 \iff B = 5p^2t + 2pqt^4 - 2q^2 - 7qp^2 = 0$. Eliminating t :

$$\text{Res}_t(R, B) = p^{15}(p - 1)^6 F(p),$$

where F is the irreducible degree-15 integer polynomial ($5764801 = 7^8$ leading, $65536 = 2^{16}$ trailing):

$$F(p) = 5764801p^{15} - 47765494p^{14} + 190003135p^{13} - 486209703p^{12} + 901678743p^{11} - 1287828143p^{10} + 1464952167p^9 - 1351039522p^8 + 1017028633p^7 - 624621984p^6 + 310300032p^5 - 122238368p^4 + 36836352p^3 - 7952896p^2 + 1073408p - 65536.$$

Irreducibility: $F \bmod 23 \in \mathbb{F}_{23}[p]$ is irreducible (leading coeff $7^8 \not\equiv 0$), so F is irreducible over $\mathbb{Q}$ (Gauss). **Sturm:** F has exactly two roots in $(0, 1)$ —these are

$$a_7 \approx 0.21427352090984097 \in (1/5, 1/4), \quad b_7 \approx 0.32862767791659197 \in (1/4, 1/3)$$

(a third real root ≈ 1.3266 lies outside).

Three-sign certificate (rational samples). Isolating the unique positive root $t(p)$ of $R(p, \cdot)$ in a rational bracket (width 4×10^{-5} , R monotone) and evaluating B by exact rational interval arithmetic gives the sign of $P_{\text{curve}} - 7$ at three rational parameters spanning the three regions:

p	region	$B(p, t(p))$
$1/5$	$p < a_7$	$B \in \left[\frac{30\,116\,709\,200\,321\,381\,334\,555\,841}{31\,250\,000\,000\,000\,000\,000\,000\,000}, \frac{34\,746\,273\,526\,146\,252\,534\,263\,041}{31\,250\,000\,000\,000\,000\,000\,000\,000} \right] > 0$
$1/4$	$a_7 < p < b_7$	$B \in \left[\frac{-170\,398\,618\,635\,119\,584\,219\,204\,477}{810^{28}}, \frac{-159\,030\,250\,817\,069\,792\,431\,615\,677}{810^{28}} \right] < 0$
$1/3$	$p > b_7$	$B \in \left[\frac{130\,644\,355\,439\,570\,080\,856\,803}{263\,671\,875\,000\,000\,000\,000\,000\,000}, \frac{164\,831\,571\,916\,353\,743\,995\,603}{263\,671\,875\,000\,000\,000\,000\,000\,000} \right] > 0$

Together with $\text{sign}(P_{\text{curve}} - 7) = \text{sign } B$ on $R = 0$ and the Sturm count (exactly two $P = 7$ crossings, at a_7, b_7), this fixes $P_{\text{curve}} > 7$ on $(0, a_7) \cup (b_7, 1)$ and $P_{\text{curve}} < 7$ on (a_7, b_7) (minimum ≈ 6.951 at $p \approx 0.27$).

Inactive KKT (zeros stay zero). For S_2 to give the global minimum the inactive derivatives $D_0 = \partial_{x_0} P|_{x_0=x_5=0}$, $D_5 = \partial_{x_5} P|_{x_0=x_5=0}$ must be ≥ 0 . The curve $R = 0$ is a single branch ($\partial R/\partial t < 0$ for $t > 0$; one positive root $t(p)$ per p). Eliminating t :

$$\text{Res}_t(R, \text{num } D_0) = -p^{23}(p-1)^{11}G(p), \quad \text{Res}_t(R, \text{num } D_5) = p^{19}(p-1)^{18}G(p),$$

sharing the degree-21 factor G (listed in §4.5). **Sturm:** G has 0 roots in $(1/5, 1/3) \supset (a_7, b_7)$; its sole $(0, 1)$ -root is $\approx 0.39387 > 1/3 > b_7$. Hence D_0, D_5 do not vanish on the curve in the band. Their sign is fixed by the rational sample $p = 1/4$ (same t -isolation as above): $D_0 \in [0.29319, 0.29464] > 0$, $D_5 \in [1.04553, 1.04895] > 0$. Thus D_0, D_5 are **strictly positive on all of (a_7, b_7)** : the S_2 stationary point is a **constrained strict local minimum** of the nonnegative simplex (the inactive derivatives $D_0, D_5 > 0$ confirm the zeros x_0, x_5 stay zero, so the minimum is attained on the S_2 face, not in the full positive cone).

4.3 S_1 —full $p \in (0, 1)$ certificate

The S_1 KKT system (5 equations in the support ratios) reduces, via the cyclic-ratio / β -coordinate elimination, to a p -free real algebraic curve. The elimination ideal has two components: the **symmetric main branch** H_B (on which $x_2 = x_5$, $x_3 = x_4$) and an **asymmetric component** H_C (treated at the end of this subsection).

Completeness of the elimination —no third component (Issue 7; auditable). We record the exact resultant certificate (`code/verify_s1_elimination.py`) that the two components above are *exhaustive*. Use the path-ratio / β coordinates $\rho_1, \dots, \rho_5$ with $\rho_{i+1} = \beta_i/(\rho(1 - \beta_i))$, $(u, v, w, z) = \beta_1, \dots, \beta_4$, $\rho = q/p$. As proved in the closed-form certificate (`code/verify_hc_closedform.py`, §4.3(i)),

$$P = A/\rho_1 + B\rho_1 + C, \quad A = (1-u)(1+\rho), \quad B = \frac{(1+\rho)uvwz}{\rho^5(1-u)(1-v)(1-w)(1-z)}, \quad C = \rho(1+\rho)S,$$

with A, B, C independent of ρ_1 ($S = (1-v)(1-u)/u + \dots + (1-z)/z$). The Euler/KKT system ($\lambda = 0$, $\beta \in (0, 1)^4$) is $g_1 := \rho_1 \partial_{\rho_1} P = \rho_1(B - A/\rho_1^2) = 0$ (hence $\rho_1^2 = A/B$) and $\partial_{\beta_j} P = 0$ for $j = 1..4$. Each $L_j := B\rho_1 \partial_{\beta_j} C + B \partial_{\beta_j} A + A \partial_{\beta_j} B$ is *linear* in ρ_1 , write $L_j = a_j \rho_1 + b_j$. Elimination chain (all exact, resultants / linear cross-multiplication, no Gröbner black box, no floats):

- (*u-elimination*) cross-multiply $L_u, L_v \Rightarrow E_{uv} = (\text{prefactor}) \cdot E_3$ with $E_3 = uv^2 + uw - u - v^2 + v$; solve $E_3 = 0$ for $u = v(1-v)/((1-w) - v^2)$.
- (*substitute u*) in E_{uv}, E_{uz} , strip the common prefactor $G = \rho v^3 w z (\rho+1)^4 (v-1)^2 (v+w-1)$ (admissibility / denominator-zero boundaries) $\Rightarrow$ two ρ -free v -relations F_1, F_2 .
- (*v-elimination*) $\text{Res}_v(F_1, F_2) = w^2 z^2 (w-1)^2 H_B H_C$ —an **exact identity** (SymPy `expand(R-target)==0`; H_B as below, H_C as in the asymmetric subsection), verified with remainder-zero exact division.

On the admissible set $w, z \in (0, 1)$ the prefactors $w^2, z^2, (w-1)^2$ are nonzero ($\beta_3, \beta_4 \in (0, 1)$), so

$$\pi(V_{\text{KKT}}^{\text{adm}}) \subseteq V(H_B) \cup V(H_C),$$

i.e. **every admissible S_1 KKT point projects to one of the two resultant factors H_B, H_C ; no third projected component is possible.** (We write containment, not equality: a resultant captures *necessary* conditions —common zeros, leading-coefficient degenerations, and lifts failing the remaining KKT equations may enlarge $V(H_B) \cup V(H_C)$; see the superset remark below. This is exactly what completeness requires.)

Localization (every stripped factor is nonzero on the admissible set). The elimination divides out, in succession: ρ, u, v, w, z and $\rho+1$ (strictly positive: $\rho > 0, \beta \in (0, 1)^4, \rho+1 > 0$); $1-v, 1-w, 1-z$ (nonzero, $\beta \in (0, 1)$); and two non-obvious factors:

- $v + w - 1$. If $v + w - 1 = 0$ then $w = 1 - v$ and $v^2 + w - 1 = v^2 - v = -v(1 - v)$, so $E_3 = u(v^2 + w - 1) + v(1 - v) = (-v(1 - v))u + v(1 - v) = v(1 - v)(1 - u) = 0$ forces $u = 1$ (as $v(1 - v) > 0$), contradicting admissibility $u \in (0, 1)$. Hence $v + w - 1 \neq 0$ on admissible KKT points.
- $(1 - w) - v^2$ (*the u -recovery denominator*). If $1 - w - v^2 = 0$ then $v^2 + w - 1 = 0$, so $E_3 = u \cdot 0 + v(1 - v) = v(1 - v) > 0$, impossible for $E_3 = 0$. Hence the denominator is nonzero.

Thus every factor removed in the chain is rigorously nonzero on $V_{\text{KKT}}^{\text{adm}}$, justifying the localisation.

Forward direction is what the theorem uses (no reverse soundness is needed). The theorem requires only the **forward** containment proved above —every genuine S_1 KKT point projects into $V(H_B) \cup V(H_C)$ —together with bounds on the *whole* positive H_B -candidate graph (monotonicity, determinant, crossing portrait below) and on the whole H_C -superset (Proposition 4.3). The reverse implication (" $H_B = 0 \Rightarrow$ a genuine KKT lift exists") is **not** needed and is not claimed: a vanishing specialised resultant can also reflect leading-coefficient degeneration (a projective root at infinity), a complex common root, or a real root outside $(0, 1)$, so $H_B \mid \text{Res}_v(F_1, F_2)$ does not by itself exhibit an admissible v -root. This is harmless: even if the H_B resultant locus contained extraneous points, analysing the entire positive H_B -candidate graph is *stronger* than analysing only the true KKT subset, and the value/determinant/crossing certificates below apply to every true KKT point that does project to H_B . (For completeness we record in `code/verify_hb_exact_backsub.py` —a *non-load-bearing* record, the load-bearing forward containment being the resultant identity above —that after the $E_3 = 0$ substitution $u = v(1 - v)/((1 - w) - v^2)$, the identity $H_B \mid \text{Res}_v(F_1, F_2)$ is reconfirmed by *genuine* univariate polynomial division (`sp.div` in w , coefficients in $\mathbb{Q}[z]$, remainder 0), and the closure/ g_1 fix $\rho, r_1 > 0$. We do **not** claim the w - or z -KKT numerator L_w, L_z is a polynomial multiple of F_1, F_2 : in fact F_1 does *not* divide $L_w|_{u=\text{usol}}$ as a v -polynomial (`sp.div` remainder $\neq 0$), because L_w depends on ρ while F_1 is ρ -free; the relation between $L_w = 0$ and $F_1 = 0$ at a KKT point is mediated by the resultant (E) and r_1 -consistency, not by a direct divisibility $L_w = \text{prefactor} \cdot F_1$. An earlier `cancel(L_w/F_1)` check was tautological —defining $q := L_w/F_1$ makes $L_w - qF_1 \equiv 0$ for any nonzero F_1 —and is removed.) For H_C , the β -variety $\{H_C, E_2^u, \bar{K}, g_1\}$ is a **superset** of the true H_C stationary set (its spurious lifts violate g_4, g_5 ; `n7_s1_hc_kkt_check.py`), and the inclusion $\{V_{\text{KKT}}^{H_C}\} \subseteq \{H_C\text{-variety}\}$ is what matters, since Proposition 4.3 bounds P on the whole superset —stronger than on the true set.

On H_B introduce

$$w = \beta_3, \quad z = \beta_4, \quad H_B(w, z) = zw^2 + (1 - z^2)w + z^2 - z = 0, \quad D = 1 - z^2 + wz^2,$$

with the positive w -root $w_+(z) = (-(1 - z^2) + \sqrt{\Delta(z)})/(2z)$, $\Delta(z) = z^4 - 4z^3 + 2z^2 + 1 > 0$ on $(0, 1)$. Setting $\rho = q/p$ ($p = 1/(1 + \rho)$), the closure and stationary value are

$$\rho^7(1 - z)D^3 - wz^5 = 0, \quad P_{S_1}^{\text{stat}} = \frac{2\rho(1 + \rho)}{z} [3 - 2z - z^2 + wz(1 + z)]. \quad (4.1)$$

Monotonicity of the branch. The closure determines ρ as a function of z alone: $\rho(z)^7 = K(z) = w_+(z)z^5/((1 - z)D(z)^3)$. Since $H_w = 2zw + 1 - z^2 > 0$ on the positive- w branch (the two H_B -roots in w have opposite sign, so the positive root has positive derivative), implicit differentiation gives the exact logarithmic derivative

$$\frac{K'(z)}{K(z)} = -\frac{2L(w, z)}{wzDH_w}, \quad L(w, z) = (2z^5 - 3z^4 - z^3 - 2z^2 + 2)w - 2z^5 + 3z^4 - z^3 + 5z^2 - 5z,$$

whose denominator is strictly positive on the branch. Thus $K' = 0 \iff L = 0$. Because L is *linear* in w , its joint zeros with H_B are governed by a single resultant,

$$\text{Res}_w(H_B, L) = z(z-1)Q_7(z),$$

where Q_7 is exactly the determinant polynomial below (unique $(0, 1)$ -root $z_7 \approx 0.87618$). The unique candidate $L = 0$ lift at z_7 is $w_L(z_7) = -b(z_7)/a(z_7)$; rational interval arithmetic gives $w_L(z_7) \in [-0.50859, -0.50858] < 0$, whereas the branch root is $w_+(z_7) \approx 0.24345 > 0$, so $L \neq 0$ on the positive- w branch. A single sign sample fixes the sign: $L(w_+(1/2), 1/2) = (-35 + 5\sqrt{17})/16 < 0$ (as $\sqrt{17} < 7$). Hence $L < 0$ throughout, so $K'(z) > 0$ and $\rho(z)$ is **strictly increasing**; equivalently $z(p)$ is strictly decreasing and the positive- w branch is a graph over $p \in (0, 1)$.

Determinant sign. The reduced Hessian determinant on H_B factors as

$$\text{Res}_w(H_B, \text{num_red}) = 4z(z-1)^9 Q_5(z) Q_7(z), \quad Q_5 = 2z^5 + 2z^3 - 2z^2 - 1, \quad Q_7 = 8z^7 - 24z^6 + 20z^5 - 9z^4 + 30z^3 -$$

Sturm: Q_5 has a unique root $z_0 \approx 0.89756$ in $(0, 1)$; Q_7 has a unique root $z_7 \approx 0.87618$ in $(0, 1)$. Since $\text{num_red} = P_v(z)w + Q_v(z)$ is *linear* in w , the determinant vanishes at a single w -value $w_{\det} = -Q_v/P_v$. At z_7 , rational interval arithmetic gives $w_{\det}(z_7) \in [-0.866, -0.248] < 0$, whereas the branch root is $w_+(z_7) \approx 0.243 > 0$; thus z_7 is **not** a determinant zero on the positive- w branch. At z_0 , $w_{\det} = w_+(z_0) > 0$ (the genuine transition). Hence on the positive- w branch $\det H_{S_1} = 0 \iff z = z_0$. Sign samples (rational interval, $D > 0$):

$$z = \frac{17}{20}, \frac{7}{8} (< z_0) : \det > 0; \quad z = \frac{9}{10}, \frac{19}{20} (> z_0) : \det < 0.$$

So $\det H_{S_1} < 0$ for $z > z_0$ and $\det > 0$ for $z < z_0$. **Load-bearing use:** only the $\det < 0$ side is rigorous for the theorem —a negative determinant of a 5×5 symmetric Hessian forces an *odd* (hence ≥ 1) number of negative eigenvalues, so the stationary point is a **saddle** for $z > z_0$; this is what defers the in-band infimum to the boundary ($\partial S_1 = S_2 \cup S_3$) below. The $\det > 0$ side ($z < z_0$) is *not* used as a Morse certificate: $\det > 0$ permits 0, 2, or 4 negative eigenvalues and does not by itself prove a local minimum; the out-of-band conclusion for $p > p_0$ instead rests directly on the stationary *value* $P_{S_1}^{\text{stat}} > 7$ (Proposition 4.2) together with Proposition 4.3 and the boundary, independent of Hessian index (see "Conclusion for H_B " below). By monotonicity $z > z_0 \iff p < p_0$ and $z < z_0 \iff p > p_0$, where $p_0 = 1/(1 + \rho(z_0)) = 0.388528131361\dots$ and rational interval arithmetic gives

$$p_0 \in (\frac{3}{8}, \frac{2}{5}) \subset (\frac{1}{3}, \frac{2}{5}) \implies \rho_0 := \rho(z_0) \in (\frac{3}{2}, 2). \quad (4.2)$$

Numerical remark (non-load-bearing). A finite-difference inertia trace (`n7_s1_hc_correct_morse_trace.py`-style; see §5) over the branch finds Morse index 1 for $p < p_0$ (one negative eigenvalue, consistent with the saddle above) and index 0 for $p > p_0$; this is an independent numerical corroboration, not a certificate the theorem relies on.

Crossing certificate (region $p > p_0$). From (4.1), $P_{S_1}^{\text{stat}} = 7$ is linear in w ; solving for w , substituting into H_B and the closure, and taking Res_z of the two resulting bivariate polynomials $A(z, \rho)$ ($\deg_z 4, \deg_\rho 4$), $B(z, \rho)$ ($\deg_z 8, \deg_\rho 11$) yields

$$\text{Res}_z(A, B) = 896\rho^{13}(\rho+1)^7(8\rho^2+8\rho+7)^6\Phi_{35}(\rho),$$

where the first three factors are strictly positive for $\rho > 0$ and Φ_{35} is the degree-35 polynomial (listed in §5). **Sturm:** Φ_{35} has exactly two positive roots,

$$\rho_1 = 2.3312465633 \dots \in (2, \frac{5}{2}), \quad \rho_2 = 3.2394376235 \dots \in (3, \frac{7}{2}),$$

both $> 2 > \rho_0$. Hence every $P_{S_1}^{\text{stat}} = 7$ crossing has $\rho > \rho_0$, i.e. $z > z_0$ (saddle region), and corresponds to

$$p_1 = \frac{1}{1+\rho_1} = 0.300187927 \dots, \quad p_2 = \frac{1}{1+\rho_2} = 0.235880343 \dots, \quad \{p_1, p_2\} \subset (a_7, b_7).$$

By monotonicity the branch over p is split by $\{p_2, p_1\}$ into three intervals. Since the resultant has no zero except at ρ_1, ρ_2 , on each open interval the sign of $P_{S_1}^{\text{stat}} - 7$ is constant; three rigorous interval samples (Krawczyk-unique z , native outward-rounded $\sqrt{\cdot}$) fix it:

$$P_{S_1}^{\text{stat}}(p) > 7 \text{ on } (0, p_2) \cup (p_1, 1), \quad P_{S_1}^{\text{stat}}(p) < 7 \text{ on } (p_2, p_1) \subset (a_7, b_7),$$

with $P_{S_1}^{\text{stat}}(2/5) \in [7.157602, 7.157603] > 7$ ($\rho = 3/2 \in (0, \rho_1)$), $P_{S_1}^{\text{stat}}(1/4) \in [6.989976, 6.989977] < 7$ ($\rho = 3 \in (\rho_1, \rho_2)$), $P_{S_1}^{\text{stat}}(1/5) \in [7.059316, 7.059317] > 7$ ($\rho = 4 \in (\rho_2, \infty)$). This is formalised as Proposition 4.2 below.

Proposition 4.2 (out-of-band S_1 ; in particular $p > p_0$). *On the symmetric branch H_B , the only $p \in (0, 1)$ at which $P_{S_1}^{\text{stat}} = 7$ are $p_1 = 0.300187927 \dots$ and $p_2 = 0.235880343 \dots$, both lying in (a_7, b_7) . Consequently*

$$P_{S_1}^{\text{stat}}(p) > 7 \text{ on } (0, p_2) \cup (p_1, 1) \supset (0, a_7] \cup [b_7, 1), \quad P_{S_1}^{\text{stat}}(p) < 7 \text{ only on } (p_2, p_1) \subset (a_7, b_7).$$

In particular, for every $p > p_0$ (where $p_0 \in (\frac{3}{8}, \frac{2}{5}) \subset (\frac{1}{3}, 1)$ is the unique Hessian-determinant transition, by (4.2)), $P_{S_1}^{\text{stat}}(p) > 7$.

Proof. The three certificates above are mutually independent and each is rigorous. (a) *Monotonicity:* $\text{Res}_w(H_B, L) = z(z-1)Q_7(z)$ and the sign analysis $w_L(z_7) < 0 < w_+(z_7)$, $L(w_+(1/2), 1/2) < 0$ give $K'(z) > 0$, so $\rho(z)$ is strictly increasing and the positive- w branch is a single graph over $p \in (0, 1)$. (b) *Crossing set:* $\text{Res}_z(A, B) = 896\rho^{13}(\rho+1)^7(8\rho^2+8\rho+7)^6\Phi_{35}(\rho)$ with the first three factors strictly positive for $\rho > 0$; Sturm gives Φ_{35} exactly two positive roots $\rho_1 \in (2, 5/2)$, $\rho_2 \in (3, 7/2)$, both $> \rho_0$ (since $\rho_0 \in (3/2, 2)$ by (4.2)), i.e. $p_1, p_2 < 1/3 < b_7$. Thus no crossing has $\rho \leq \rho_0$ (equivalently $p \geq p_0$): $P_{S_1}^{\text{stat}} \neq 7$ on $[p_0, 1)$. (c) *Sign of the three components.* Since the resultant has no zero except at ρ_1, ρ_2 , on each open interval $(0, \rho_1)$, (ρ_1, ρ_2) , (ρ_2, ∞) the sign of $P_{S_1}^{\text{stat}} - 7$ is constant; three rigorous interval samples (Krawczyk-unique z on the closure, native outward-rounded $\sqrt{\cdot}$, `mpmath.iv`; `_verify_s1_three_samples.py`) fix it:

component	sample (ρ, p)	$P_{S_1}^{\text{stat}}$
$(0, \rho_1) \leftrightarrow (p_1, 1)$	$(3/2, 2/5)$	$[7.157602, 7.157603] > 7$
$(\rho_1, \rho_2) \leftrightarrow (p_2, p_1)$	$(3, 1/4)$	$[6.989976, 6.989977] < 7$
$(\rho_2, \infty) \leftrightarrow (0, p_2)$	$(4, 1/5)$	$[7.059316, 7.059317] > 7$

In particular $[p_0, 1) \subset (p_1, 1)$ has $P_{S_1}^{\text{stat}} > 7$. (Corroboration: Φ_{35} is squarefree, $\gcd(\Phi_{35}, \Phi'_{35}) = 1$, so both crossings are simple — `_verify_phi35_squarefree.py`.) The inclusions $\{p_1, p_2\} \subset (a_7, b_7)$ follow from exact Sturm isolation of all four roots: $a_7 \in (0.214, 0.215)$, $b_7 \in (0.328, 0.329)$ (the two $(0, 1)$ -roots of F), $p_2 \in (0.235, 0.236)$, $p_1 \in (0.300, 0.301)$ (from $\rho_1 \in (2, 5/2)$, $\rho_2 \in (3, 7/2)$); hence $a_7 < 0.215 < 0.235 < p_2 < 0.236 < 0.328 < b_7$ and $a_7 < 0.215 < 0.300 < p_1 < 0.301 < 0.328 < b_7$ (`n7_s1_crossing_resultant.py`, `_verify_crossings_in_band.py`). $\square$

Conclusion for H_B . Combining the certificates (and recalling that $\inf_{S_1} P = \min\{\text{interior local-min values, } \inf_{\partial S_1} P\}$, with interior local minima a subset of the stationary points $H_B \cup H_C$):

- *In-band* $p \in (a_7, b_7) \subset (0, p_0)$: $\det H_{S_1} < 0 \Rightarrow$ saddle, so the H_B stationary point is *not* a local minimum and does not supply the infimum; $\inf_{S_1} P$ is attained on $\partial S_1 = S_2 \cup S_3$, where $S_2 < 7$ (the failure, §4.2) and $S_3 > 7$ (§4.1). Hence $\inf_{S_1} P = \inf_{S_2} P < 7$ in-band.
- *Out-of-band* $p \notin (a_7, b_7)$: every interior stationary value is > 7 —the H_B value by Proposition 4.2 ($P_{S_1}^{\text{stat}} > 7$ on $(0, a_7] \cup [b_7, 1)$), the H_C value by Proposition 4.3 ($\geq L_C > 7$) — and $\inf_{\partial S_1} P = \min\{\inf_{S_2} P, \inf_{S_3} P\} \geq 7$ out-of-band (§4.2, §4.1). Therefore $\inf_{S_1} P \geq 7$ out-of-band, *regardless* of whether the H_B stationary point is a local minimum or a saddle (its value is > 7 either way).

Asymmetric component H_C . H_C is the cubic-in- w factor of the elimination ideal,

$$H_C(w, z) = zw^3 + z^3w^2 - zw^2 + z^4w - 3z^3w + 2z^2w + zw - w - z^4 + 3z^3 - 3z^2 + z,$$

appearing as the second factor of the exact completeness identity $\text{Res}_v(F_1, F_2) = w^2z^2(w - 1)^2H_BH_C$ recorded above. Its admissible positive lifts (solving $H_C = 0$ and the two remaining p -free β -recurrences, then the closure) exist only for $z \in (\zeta, 1)$, ζ the unique $(0, 1)$ -root of $z^3 + 2z^2 - z - 1$, and correspond to $p \in (0.3885, 0.3930) \subset (b_7, 1)$ (out-of-band). The rigorous bound on H_C is **not** a pointwise Morse classification but a validated cover of the *entire admissible H_C -superset* (Proposition 4.3 below): every admissible point has $P \geq L_C = 141/20 > 7$, certified by interval-Newton uniqueness, strict admissibility, and a mean-value P -bound, independently re-verified on two interval backends (mpmath.iv and Arb). The would-be second lift branch near $p \approx 0.52$ gives KKT residual > 2 and is not a stationary point (script `n7_s1_hc_kkt_check.py`), so the β -variety is a genuine *superset* of the true H_C stationary set; bounding P on the superset is stronger than on the true set.

Numerical remark (representative samples, not a rigorous count). A finite-difference trace over $z \in \{0.80, \dots, 0.99\}$ (reconstructing x from the 1D free-height equation $g_1 = 0$ and checking all five KKT residuals $\sim 10^{-7}$) finds eight representative real H_C lifts, all with $P > 7$ (smallest sampled $P \approx 7.1334$ near $z \approx 0.90$); these are illustrative samples of the validated arc, not a rigorous enumeration or Hessian-inertia certificate. The theorem relies solely on the superset bound $P \geq L_C > 7$.

Proposition 4.3 (validated exclusion of H_C). *Every admissible point of the reduced H_C -superset satisfies $P \geq L_C > 7$, where $L_C = \frac{141}{20} = 7.05$ is an explicit rational. Consequently every true H_C stationary point has value > 7 .*

Proof. As noted above, the reduced system $H_C = 0 \wedge E_2^u = 0 \wedge \bar{K} = 0 \wedge g_1 = 0$ (closure $K = uvwz^3a_5^2/((1-v)(1-w)(1-z)^3)$, u eliminated by E_2^u) is a *superset* of the true H_C stationary set (its spurious branch violates g_4, g_5), so bounding P on the whole admissible superset is *stronger* than on the true set.

(i) Closed form (exact). Write the S_1 path ratios $\rho_1, \dots, \rho_5$ ($x = (0, 1, \rho_1, \rho_1\rho_2, \dots, \rho_1 \cdots \rho_5)$) with $\rho_{i+1} = \beta_i/(\rho(1 - \beta_i))$ ($\beta_i \in (0, 1)$, $\rho = q/p$, $p = 1/(1 + \rho)$), and set $(u, v, w, z) = (\beta_1, \beta_2, \beta_3, \beta_4)$. Expanding $P = \sum_{i=1}^4 \frac{1}{\rho_i(p+q\rho_{i+1})} + \frac{1}{p\rho_5} + \frac{\prod \rho_i}{q}$ in these coordinates gives the **rational identity** (verified by exact SymPy expansion, numerator identically zero — `verify_hc_closedform.py`)

$$P = \frac{A}{\rho_1} + B\rho_1 + C,$$

with

$$A = (1-u)(1+\rho), \quad B = \frac{(1+\rho)uvwz}{\rho^5(1-u)(1-v)(1-w)(1-z)}, \quad C = \rho(1+\rho)S, \quad S = \frac{(1-v)(1-u)}{u} + \frac{(1-w)(1-v)}{v} + \frac{(1-z)(1-w)}{w} + \frac{1-z}{z},$$

where A, B, C are **independent of ρ_1** . The first KKT equation is $g_1 := \rho_1 \partial P / \partial \rho_1 = \rho_1(B - A/\rho_1^2)$, so on the admissible set ($\rho_1 > 0$)

$$g_1 = 0 \iff \rho_1^2 = A/B \iff \rho_1 = +\sqrt{A/B} \quad (A, B > 0 \text{ on the admissible set}).$$

Substituting $\rho_1 = \sqrt{A/B}$: $A/\rho_1 + B\rho_1 = A\sqrt{B/A} + B\sqrt{A/B} = \sqrt{AB} + \sqrt{AB}$, whence

$$\boxed{P = C + 2\sqrt{AB}}$$

as an **exact algebraic identity** on the H_C stationary set —no numerical approximation is involved (the squared check $(A/\rho_1 + B\rho_1)^2|_{\rho_1^2=A/B} = 4AB$ is also verified identically). Numerically $C \in (3.7, 5.2) < 7$ throughout and $2\sqrt{AB} \in (2.95, 3.47)$. On the admissible set all denominators and radicands are strictly positive, so P is C^1 there.

(ii) Desingularisation. The admissible arc terminates at $(w, z) = (0, 1)$, where $H_C(w, 1) = w^3$ and $\partial_w H_C, \partial_z H_C \rightarrow 0$ (triple root). With $s = 1 - z$, $c = w/s$ the factor divides out: $H_C(cs, 1-s) = s^3 G(c, s)$, $G = (1-s)c^3 + (1-s)(s-2)c^2 + (s^2-s-1)c + (1-s)$, **regular** at $s = 0$ ($G(c, 0) = c^3 - 2c^2 - c + 1$ has the arc root $c_0 \approx 2.24698$, $\partial_c G(c_0, 0) \approx 5.15 \neq 0$); the closure and E_2^u lose their s -power degeneracy and give regular reduced equations $G(c, s) = 0$, $E_2^{\text{red}}(v, c, s) = 0$.

(iii) Validated cover (completeness). The admissible-lift set changes only at finitely many z -events, collected by resultants in `n7_s1_hc_critical_events.py`: root multiplicity (c -double E_1 , v -double E_2) and admissibility-boundary crossings ($w=1$ E_3 , $v=0$ E_4 , $v=1$ E_5 , $u=1$ E_6 , $a_5=0$ E_7 , u -denom= 0 E_8 , $K=0$ E_9). Their $(0, 1)$ -roots give five critical z -values $\{0.295289, 0.365689, 0.569840, 0.571538, 0.801938\}$, mapped to $s = 1 - z$, partitioning $(0, 1)$ into six event-free cells. On each cell no lift crosses a boundary, so the admissible-lift count is constant. **Completeness certificate** (`n7_s1_hc_completeness_exact.py` —an *exact algebraic* certificate: no floating-point root recovery, no residual threshold, no count-only comparison). The five critical z -events $E_1 - E_9$ are kept as exact `CRootOf` algebraic numbers with rational isolation intervals (never converted to float); the rational boundary between two consecutive critical roots is proposed by a high-precision evalf seed but *certified* by the exact `CRootOf`-vs-`Rational` comparison to lie strictly between them, so each cell is genuinely event-free. At a rational midpoint s_0 of each cell: (a) the resultant $R(v) = \text{Res}_c(G, E_2^{\text{red}})|_{s_0}$ (rational) is exact-Sturm-counted for *distinct* roots in the open interval $(0, 1)$ and each root α is isolated as an exact `CRootOf` (Sturm count = `CRootOf` count, asserted); (b) for each v -root α , the exact c -roots β_j of $G(c, s_0) = 0$ (`CRootOf`) are tested for a *genuine common zero* by the exact algebraic-number identity $E_2^{\text{red}}(\alpha, \beta_j, s_0) \stackrel{?}{=} 0$ (`nsimplify+simpify`, no `nroots`, no 10^{-6} residual), ruling out spurious resultant roots (leading-coefficient drop / complex / out-of-interval); (c) admissibility ($\beta > 0$, $0 < w < 1$, $0 < v < 1$, $0 < u < 1$, $a_5 > 0$, $K > 0$) is certified by *exact algebraic sign tests* (`CRootOf/Rational` and algebraic-expression > 0), with no 10^{-9} threshold; (d) each certified admissible lift (α, β) is matched to a cover box by *exact* interval containment (s_0 in the box's rational s -range, $c_{\text{lo}} < \beta < c_{\text{hi}}$, $v_{\text{lo}} < \alpha < v_{\text{hi}}$, all exact algebraic-vs-rational), and this is a **bijection** —each lift lies in exactly one box *and* each box at s_0 contains exactly one lift (uniqueness in both directions, not a count comparison). In every cell the bijective match holds and the Sturm, `CRootOf`, and admissible-lift counts agree (`all_cells_ok=True`); the five event fibers are certified, by the same exact `CRootOf`-vs-`Rational` containment, to lie strictly inside a Krawczyk-certified cover piece (`all_event_fibers_covered=True`). Since the resultant captures *all* common zeros and the cell is event-free (constant admissible-lift cardinality), the midpoint bijective match covers every lift on the whole cell —no admissible lift is missed.

(iv) Validated cover (bounding). Each admissible lift is continued over its s -cell by interval-Newton (Krawczyk) contraction on G (in c) and E_2^{red} (in v) with factored derivative evaluators: $N(W) \subset \text{int } W$ certifies a *unique* root in the box. The stationary value is bounded over the 4-box (v, c, s, ρ) by the **mean-value form** $P(\text{box}) \subseteq P(\text{mid}) + \sum_i (\partial P / \partial x_i)(\text{box}) \cdot (\text{box}_i - \text{mid}_i)$, the four partials being closed-form factored (no symbolic simplification, avoiding interval blow-up from fractional powers with sign-changing bases); inputs are outward-rounded. The s -partition is adaptively bisected whenever the lower bound falls below $L_C = 141/20$.

(v) **Result.** The five event fibers are certified at point- s (each carries one admissible lift, all with $P \geq 7.926$); the endpoint $s = 1$ ($z = 0$) is empty ($G(c, 1) = -c$), and the endpoint $s = 0$ ($z = 1$, the degenerate limit $w \rightarrow 0$, $K \rightarrow 0$, not admissible) is **covered, not merely approached**: the desingularisation $G(c, s)$ is regular at $s = 0$ ($G(c, 0) = c^3 - 2c^2 - c + 1$, simple arc root $c_0 \approx 2.24698$), so the cover includes two pieces with closed lower endpoint $s_{10} = 0$ (namely $s \in [0, 9.67 \times 10^{-5}]$, real-arc and a second branch), certified with $P_{MV} \geq 7.1139$ and ≥ 8.2036 respectively —both > 7 , so the whole interval $(0, 10^{-7}) \subset (0, 9.67 \times 10^{-5})$ is rigorously covered (real-arc limit $P \rightarrow 7.146684 > 7$). The canonical cover (2604 rational lift-records, produced by `n7_s1_hc_arb_refine.py`) is certified by the **formal validated-numeric backend** `n7_s1_hc_arb_checker.py` (Arb / `python-flint`, prec 150): it re-reads the dumped boxes and re-derives, per piece, Krawczyk uniqueness, strict admissibility, and the mean-value bound in a hand-written lo/hi interval layer (monotone-endpoint enclosures for $\sqrt{\cdot}$ and $\rho^{1/7}$, every operation rigorously outward-rounded, dependency reduced by factored evaluators), verifying $P_{MV} > L_C$ for **all** 2604/2604 pieces; the minimum of these certified piecewise lower bounds is $\min_j P_{MV}^{(j)} = 7.050005023 \dots > 141/20$. (This is a *rigorous lower bound on P* over the superset, obtained as the minimum of the per-piece mean-value lower bounds; we make no claim that it equals the exact infimum of P on the H_C -superset —only that it exceeds $141/20 > 7$, which is all the proof requires.) As an independent defence-in-depth, the cover is **re-verified by a second interval backend** —`n7_s1_hc_cover_checker.py` (`mpmath.iv`, native outward-rounded `sqrt`) —which independently re-derives the same Krawczyk inclusion, admissibility, and mean-value bound and confirms $P_{MV} > L_C$ for **all** 2604/2604 pieces. (The few tight real-arc pieces, whose mean-value lower bound lay within 0.002 of L_C and were widened below it by interval dependency, were resolved by s -bisection —`n7_s1_hc_arb_refine.py` —not by any relaxation of rigor; both backends are fully rigorous and outward-directed.) Hence

$$P \geq L_C = \frac{141}{20} = 7.05 > 7 \quad \text{on the entire admissible } H_C\text{-superset,}$$

and in particular on every true H_C stationary point. (Full per-piece data and checkers are contained in the archived proof artifact described in §5.) $\square$

Thus every positive S_1 stationary point either has $P \geq 7$, or lies in (a_7, b_7) with a non-PSD Hessian: on H_B by Proposition 4.2 (out-of-band $P_{S_1}^{\text{stat}} > 7$ for $p > p_0$; saddle for $p < p_0$, deferring to the boundary), on H_C by Proposition 4.3 ($P \geq L_C > 7$). Consequently $\inf_{S_1} P \geq 7$ for $p \notin (a_7, b_7)$ and $= \inf_{S_2} P < 7$ for $p \in (a_7, b_7)$.

4.4 S_0 —Nowosad–Yamagami

The uniform point is handled by Lemma 2.2: the Hessian eigenvalues $\mu_k = 2(1 - \cos \theta_k)[2q \cos \theta_k + 2p^2 - 3p + 2]$ ($\theta_k = 2\pi k/7$) are strictly positive because $\Delta_7 < 0$ (§4.1), and S is invertible (no Fourier eigenvalue $\lambda_k = p\omega^k + q\omega^{2k}$ vanishes for odd n). Hence **1** is the **unique interior local-minimum ray** on S_0 for every $p \in (0, 1)$, with value 7. The face infimum is *not* 7 pointwise: by boundary inheritance (§4.5) $\inf_{S_0} P = \min\{7, \inf_{S_1} P, \inf_{S_2} P, \inf_{S_3} P, \inf_{S_4} P\}$, so in the band (a_7, b_7) where $\inf_{S_2} P < 7$, one has $\inf_{S_0} P < 7$ —the failure is inherited from the S_2 boundary stratum, not contradicted by anything at the uniform point.

4.5 Conclusion of Theorem 2

Assembling the five face certificates for every $p \in (0, 1)$ by **boundary inheritance** ($\inf_{S_0} P = \min\{7, \inf_{S_1} P, \inf_{S_2} P, \inf_{S_3} P, \inf_{S_4} P\}$, and each $\inf_{S_k}$ is itself either an interior stationary value or inherited from ∂S_k): S_0 contributes the attained uniform value 7 (unique interior local min, Lemma 2.2, §4.4); S_4 gives $P \geq \frac{16}{3^{3/4}} > 7$ (§4.1); S_3 gives $P > 8$ ($p > 1/2$) or defers to S_4 ($p \leq 1/2$) (§4.1); S_1 gives $\inf_{S_1} P \geq 7$ out-of-band and $= \inf_{S_2} P$ in-band (Propositions 4.2–4.3, §4.3); and S_2 gives $P_{S_2}^{\text{curve}} > 7$ out-of-band, < 7 in-band (§4.2). Since S_4 is terminal and every boundary recursion lands in one of these faces, the only face that can drop below 7 is S_2 , hence

$$m_7(p) = \min\{7, P_{S_2}^{\text{curve}}(p)\}, \quad H_7 = (0, a_7] \cup [b_7, 1).$$

The degree-21 polynomial

$$\begin{aligned} G(p) = & 13p^{21} - 238p^{20} + 1841p^{19} - 8295p^{18} + 25690p^{17} - 60200p^{16} + 114261p^{15} \\ & - 184911p^{14} + 263550p^{13} - 333970p^{12} + 372736p^{11} - 359996p^{10} + 295750p^9 - 203770p^8 \\ & + 116300p^7 - 54264p^6 + 20349p^5 - 5985p^4 + 1330p^3 - 210p^2 + 21p - 1 \end{aligned}$$

certifies the inactive KKT (Sturm: 0 roots in $(1/5, 1/3) \supset (a_7, b_7)$). ■

Remark (sharpness vs. $n = 5$). The Tuan–Thuong sufficient interval $I_{\text{TT}} = [(5 - \sqrt{5})/10, (5 + \sqrt{5})/10] \approx [0.2764, 0.7236]$ (a strict subset of $H_5 = (0, 1)$, Theorem 1) does **not** extend to $n = 7$: since $a_7 \approx 0.2143 < (5 - \sqrt{5})/10 \approx 0.2764 < b_7 \approx 0.3286$, the whole interval $((5 - \sqrt{5})/10, b_7)$ lies in $H_5 = (0, 1)$ but inside the $n = 7$ failure band (a_7, b_7) . The transition from “ $n = 5$ safe everywhere” to “ $n = 7$ first failure band” is boundary-induced, exactly as the gap-concentration picture predicts (the $L = 5$ one-gap face S_2 first drops below 7).

5 Computational supplement

Every certificate is reproduced by independent Python/SymPy scripts using **exact rational arithmetic** (`Poly.resultant`, `Poly.count_roots`/Sturm sequences, exact `exquo` division with zero remainder, finite-field irreducibility). Sign and inequality certificates use **rational interval arithmetic** (monomial-wise bounds, or `mpmath.iv` with native outward-rounded `sqrt`/ n th-root). The H_C cover uses Arb / `python-flint` (lo/hi interval layer with outward-rounded $\sqrt{\cdot}$ and n th-root, factored dependency-reduced evaluators) as the **formal validated-numerics backend**, and is **independently re-verified** by a second backend (`mpmath.iv`, native outward-rounded `sqrt`); both backends are fully rigorous (outward-directed). Exact-algebra scripts and high-precision sanity checks are kept in separate files. Software: Python 3.13.9, SymPy 1.14, mpmath 1.3, python-flint 0.9 (Arb), NumPy/SciPy (sanity only); versions pinned in `code/requirements_lock.txt`.

The S_1 crossing polynomial $\Phi_{35}(\rho)$ (§4.3), leading coeff 2^{18} , trailing 191102976:

$$\begin{aligned} (c_0, \dots, c_{35}) = & (191102976, 2193551360, 11883053056, 41692980224, 107322284272, 221073893824, \\ & 410624553696, 823732168256, 1952186204080, 4815489002744, 10772897763040, 20701541228760, \\ & 33774085926224, 46846168825232, 55494399759599, 56982453166940, 51848980402968, 43526037225048, \\ & 34334035751596, 25279161341952, 15843550970240, 7848268705920, 2306869918304, 88546441088, \\ & -543801267968, -327985403648, -166554596992, -34693234688, -9171843072, 2185738240, \\ & 547314432, 444107776, 57874432, 19453952, 1211392, 262144). \end{aligned}$$

Sturm sign-variation counts for Φ_{35} : $V(0) - V(\infty) = 2$; isolation $V(0) - V(2) = 0$, $V(2) - V(5/2) = 1$, $V(5/2) - V(3) = 0$, $V(3) - V(7/2) = 1$, $V(7/2) - V(\infty) = 0$.

Region- $p > p_0$ certificate (Proposition 4.2). Three independent certificates, each rigorous: (i) *monotonicity* — $\text{Res}_w(H_B, L) = z(z - 1)Q_7(z)$, rational-interval $w_L(z_7) \in [-0.50859, -0.50858] < 0 < w_+(z_7)$ and $L(w_+(1/2), 1/2) = (-35 + 5\sqrt{17})/16 < 0$, giving $\rho(z)$ strictly increasing (`n7_s1_rho_monotone_Lcert.py`); (ii) *crossing set* — Φ_{35} has exactly two positive roots $\rho_1 \in (2, 5/2)$, $\rho_2 \in (3, 7/2)$, both $> \rho_0 \in (3/2, 2)$, so $P_{S_1}^{\text{stat}} \neq 7$ on $[p_0, 1)$ (`n7_s1_crossing_resultant.py`); (iii) *three interval samples* — on each open ρ -interval (sign-constant, since the resultant has no other zero) a rigorous sample fixes the sign: $P_{S_1}^{\text{stat}}(2/5) \in [7.157602, 7.157603] > 7$ ($\rho = 3/2 \in (0, \rho_1)$), $P_{S_1}^{\text{stat}}(1/4) \in [6.989976, 6.989977] < 7$ ($\rho = 3 \in (\rho_1, \rho_2)$), $P_{S_1}^{\text{stat}}(1/5) \in [7.059316, 7.059317] > 7$ ($\rho = 4 \in (\rho_2, \infty)$), each by Krawczyk-unique z -isolation on the closure with native outward-rounded `mpmath.iv sqrt`

(`_verify_s1_three_samples.py`). (Corroboration: Φ_{35} is squarefree, $\gcd(\Phi_{35}, \Phi'_{35}) = 1$, so the crossings are simple — `_verify_phi35_squarefree.py`.) The inclusion $\{p_1, p_2\} \subset (a_7, b_7)$ is certified by exact Sturm isolation of all four roots: $a_7 \in (0.214, 0.215)$, $b_7 \in (0.328, 0.329)$, $p_2 \in (0.235, 0.236)$, $p_1 \in (0.300, 0.301)$ (`_verify_crossings_in_band.py`).

Files (in code/):

- $n = 5$. `n5_s1_certificate.py` —stationary curve $q^3a^6 - p^3a^2 - p^2q = 0$, $\text{Res}_a = r^{17}Q_{11}(r)$ (degree-11, all-positive), $M(1) > 0$. `n5_s1_m1_interval.py` — $a(1) \in (1.150, 1.151)$, $M(1) \in [0.5662, 0.5800] > \frac{1}{2}$ by rational interval arithmetic.
- $n = 7$, S_2 . `n7_resultant.py/n7_roots.py` — F (degree 15), $F \bmod 23$ irreducible (finite-field cert), exactly two $(0,1)$ -roots. `n7_s2_rational_signs.py` —three-sign rational-interval samples of B at $p = 1/5, 1/4, 1/3$ (width- 4×10^{-5} t -brackets), $D_0, D_5 > 0$ at $p = 1/4$. `n7_inactive_sturm.py` — G (degree 21), Sturm (0 roots in $(1/5, 1/3)$).
- $n = 7$, S_1 . `n7_s1_crossing_resultant.py` —builds A, B , computes $\text{Res}_z = 896\rho^{13}(\rho+1)^7(8\rho^2+8\rho+7)^6\Phi_{35}$ (exact exquo, remainder 0; all 36 coefficients match), Sturm counts. `_verify_crossings_in_band.py` —exact Sturm isolation of a_7, b_7, p_1, p_2 certifying $\{p_1, p_2\} \subset (a_7, b_7)$ and $p_1, p_2 < 1/3 < p_0$. `_verify_phi35_squarefree.py` — $\gcd(\Phi_{35}, \Phi'_{35}) = 1$ (squarefree, so both crossings are simple) + Sturm zero-counts; **corroboration only** —the three-segment sign portrait itself is certified by `_verify_s1_three_samples.py` (three rigorous interval samples, not by squarefree alternation). `n7_s1_rigorous_certs.py` —`mpmath.iv`: $P_{S_1}^{\text{stat}}(2/5) > 7$; det sign at $z = 17/20, 7/8, 9/10, 19/20$; $w_{\text{det}}(z_7) < 0$; $p_0 \in (3/8, 2/5)$. `_verify_s1_three_samples.py` —three rigorous interval samples ($p = 2/5, 1/4, 1/5$) fixing the H_B three-segment sign portrait (Krawczyk-unique z , native outward-rounded `sqrt`). `n7_s1_rho_monotone_check.py` — $F'(z) \neq 0$ on $(0, 1)$ (Sturm on $G = A^2 - B^2\Delta$, deg 18; sole $(0,1)$ -root z_7 extraneous). `n7_s1_rho_monotone_Lcert.py` — L -certificate: $\text{Res}_w(H_B, L) = z(z-1)Q_7(z)$, rational-interval $w_L(z_7) < 0 < w_+(z_7)$, $L(w_+(1/2), 1/2) < 0$, hence $\rho(z)$ strictly increasing. `n7_s1_correct_morse_trace.py` —corrected finite-difference Hessian (diag/off formulas, three step sizes), Morse index. `n7_s1_all_branches_correct_hess.py` —all-branch enumeration. `n7_s1_hc_definitive_trace.py` — H_C trace (8 admissible points, all $P > 7$). `n7_s1_hc_kkt_check.py` —all-five-KKT-residual check at H_C lifts; proves the β -variety is a *superset* (spurious branch fails g_4, g_5). `n7_s1_hc_Cstruct.py` —probes $P = C + 2\sqrt{AB}$ ($C < 7$, $2\sqrt{AB} \in (2.95, 3.47)$). `n7_s1_hc_critical_events.py` —enumerates the five critical z -events (resultants E_1 – E_9 , sizes ≤ 8) partitioning $(0, 1)$. `n7_s1_hc_rigorous_cert.py` —the full rigorous H_C certificate: (c, s) desingularisation, Krawczyk root isolation on G and E_2^{red} , mean-value-form P bound, global $P_C \geq L_C = 141/20$ (`mpmath.iv`, native outward-rounded `sqrt`, `ivprec` 130). `n7_s1_hc_completeness_exact.py` —the **exact algebraic** completeness certificate (load-bearing for *exhaustiveness*; the two interval checkers certify only box *validity*, not that every admissible lift is *enumerated*): critical z -events as exact `CRootOf` (no float), rational cell boundaries verified by exact `CRootOf`-vs-`Rational` comparison, per-cell exact Sturm distinct-root count = `CRootOf` count, exact algebraic common-zero test (no `nroots`, no residual threshold), exact algebraic sign-admissibility (no 10^{-9} threshold), and a **bijective** lift cover-box match (uniqueness in both directions, not count-only) on all six cells (`all_cells_ok=True`); event fibers certified inside a Krawczyk piece by exact containment (`all_event_fibers_covered=True`). `n7_s1_hc_completeness.py` (the earlier float-based version) is retained as a cross-check and is superseded by this exact script. `n7_s1_hc_cover_dump.py` —dumps the machine-checkable cover to `code/_hc_cover.json` (rational $s/c/v$ boxes, admissibility lower bounds, $L_C = 141/20$). `n7_s1_hc_arb_refine.py` —refines the few tight real-arc s -pieces (whose mean-value lower bound, before refinement, lay in $(7.0487, 7.05)$ —below L_C due to interval dependency, not because the true value is below L_C ; the minimum of the certified

piecewise lower bounds after refinement is $\min_j P_{\text{MV}}^{(j)} = 7.050005023\dots$, only $\approx 5 \times 10^{-6}$ above L_C) by s -bisection until the rigorous Arb mean-value bound clears L_C ; produces the canonical 2604-piece cover. `n7_s1_hc_cover_checker.py` —short deterministic checker (`mpmath.iv`): re-derives per piece the Krawczyk inclusion, strict admissibility, and $P_{\text{MV}} > L_C$; verifies 2604/2604 pieces. `n7_s1_hc_arb_checker.py` —**independent Arb / python-flint checker** (prec 150): a hand-written lo/hi interval layer (monotone-endpoint $\sqrt{\cdot}$, $\rho^{1/7}$; 4-product signed mul) independently re-derives Krawczyk uniqueness, admissibility, and the mean-value bound; verifies 2604/2604 pieces, with minimum certified lower bound $\min_j P_{\text{MV}}^{(j)} = 7.050005023\dots > 141/20$. `n7_s1_hc_mv_verify.py` —validity check $P_{\text{MV}} \leq P_{\text{true}}$ at sampled lifts. `s1_beta_reduction.py/verify_beta_reduction.py` — β -coordinate KKT reduction.

- **Exact symbolic verifiers (Lemma 2.2 bridge + S_1 structure; Issues 4/6/7 + GPT b3).** `verify_ny_spectral.py` —exact SymPy proof of the N-Y spectral bridge (Lemma 2.2, (NY-2)): $\mu_k = 2|\lambda_k|^2 - 2\operatorname{Re} \lambda_k \equiv 2(1 - \cos \theta_k)[2q \cos \theta_k + 2p^2 - 3p + 2]$ (identity (A)); $\operatorname{sign} \mu_k = \operatorname{sign}(|\lambda_k - \frac{1}{2}|^2 - \frac{1}{4})$, i.e. $\mu_k > 0 \Leftrightarrow \lambda_k$ outside the closed disk $|\lambda - \frac{1}{2}| \leq \frac{1}{2}$ (equivalence (B)); $\Delta_n = 4c_n^2 - 4c_n - 7 < 0$ for $n = 5, 7$ with the dangerous-mode bracket discriminant $= \Delta_n$ and leading coefficient $2 > 0$, hence every $\mu_k > 0$ (check (C)). `verify_hc_closedform.py` —exact SymPy proof of the S_1 stationary closed form $P = A/\rho_1 + B\rho_1 + C$ (rational identity, numerator $\equiv 0$), $g_1 = \rho_1(B - A/\rho_1^2)$, and the squared check $(A/\rho_1 + B\rho_1)^2|_{\rho_1^2=A/B} = 4AB$, giving the boxed exact identity $P = C + 2\sqrt{AB}$ (Issue 6; no floats). `verify_s1_elimination.py` —the exact S_1 elimination-completeness certificate (Issue 7): stepwise resultant elimination (no Gröbner black box) yielding $\operatorname{Res}_v(F_1, F_2) = w^2 z^2 (w-1)^2 H_B H_C$ as an exact identity (`expand(R-target)==0`), proving the S_1 KKT locus is **contained in** $\{H_B = 0\} \cup \{H_C = 0\}$ with no third projected component (containment, not equality —the forward direction the theorem uses); saturation prefactors $w, z, w-1$ localised away at $\beta \in (0, 1)^4$. `verify_hb_exact_backsub.py` —a **non-load-bearing** forward-construction record on H_B (the load-bearing forward containment is `verify_s1_elimination.py`); the earlier tautological `cancel(L_w/F_1)` check —defining $q := L_w/F_1$ makes $L_w - qF_1 \equiv 0$ for any nonzero F_1 —is removed. It reconfirms, by *genuine* polynomial division (`sp.div` in w , remainder 0), that $H_B \mid \operatorname{Res}_v(F_1, F_2)$ with exact quotient $R/H_B = w^2(w-1)^2 H_C$ (this script uses the unscaled F_1 ; the canonical $w^2 z^2 (w-1)^2 H_B H_C$ of identity (E), with the extra z^2 from the $-z$ -scaled F_1 , is the load-bearing form in `verify_s1_elimination.py` —same containment, scaled); and it honestly records that F_1 does **not** divide $L_w|_{u=\text{usol}}$ (L_w depends on ρ , F_1 is ρ -free), so the $L_w=0 \leftrightarrow F_1=0$ relation at a KKT point is mediated by the resultant and r_1 -consistency, not by direct divisibility. No reverse soundness ($H_B=0 \Rightarrow$ KKT lift) and no float residual are invoked.

The S_1/S_2 crossing and determinant certificates are univariate (Sylvester determinants of size ≤ 12). The H_C critical-event enumeration forms bivariate resultants (eliminate v via E_2^u , then w via H_C), but each factor has degree ≤ 3 in the eliminated variable, so every intermediate has bounded degree and the scripts run in bounded memory.

Reproducibility and archival (Issue 9). Every load-bearing artifact is committed to a self-contained local git repository with a SHA-256 manifest (`sha256_manifest.txt`, in standard HASH path form so `sha256sum -c` passes on a fresh extract) and a pinned dependency lock (`code/requirements_lock.txt`: Python 3.13.9, `sympy==1.14.0`, `mpmath==1.3.0`, `numpy==2.4.4`, `scipy==1.18.0`, `python-flint==0.9.0`). The seven scripts above (five exact SymPy —including the exact-algebraic exhaustiveness certificate `n7_s1_hc_completeness_exact.py` —and two independent interval backends) all pass on a clean checkout; a captured run is in `code/_archival_run.txt`. The 2604-piece H_C cover is published as `code/_hc_cover.json` and is re-verified bit-for-bit by both `n7_s1_hc_cover_checker.py`

(`mpmath.iv`) and `n7_s1_hc_arb_checker.py` (independent Arb/python-flint): 2604/2604 pieces, with minimum certified lower bound $\min_j P_{MV}^{(j)} = 7.050005023\dots > L_C = 141/20 > 7$ (a rigorous lower bound on P over the whole superset; we make no claim that this equals the exact infimum of P on the superset). The two interval checkers certify box *validity*; `n7_s1_hc_completeness_exact.py` certifies the orthogonal *exhaustiveness* property (every admissible lift is enumerated by a box) by exact algebra. The proof-artifact commit hash and the Zenodo DOI are recorded in `ARCHIVAL.md` (authoritative on any checkout via `git rev-parse HEAD`); full reproduction instructions are likewise in `ARCHIVAL.md`.

References

- [1] V. G. Drinfel'd, *A certain cyclic inequality*, Math. Notes **9** (1971), 68–71 (the asymptotic Shapiro cyclic constant).
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- [4] S. Yamagami, *Cyclic inequalities*, Proc. Amer. Math. Soc. **118** (1993), 521–527.
- [5] T. Ando, *A New Proof of Shapiro Inequality*, Math. Inequal. Appl. **16** (2013), 611–632.

Status. Theorems 1 and 2 are fully rigorous. Every algebraic certificate (resultant, Sturm sign-variation count, positive-coefficient factor, Hessian Morse factor, rational interval sign) is locally reconstructed and verified by exact division. The $n = 9$ case and the general odd- n conjectures are the subject of a companion paper.